

AI: An Introduction to Natural Language Processing

In the last article in this series on AI and machine learning, we developed further insights on the workings of neural networks. Subsequently, we enhanced our understanding of unsupervised learning through the use of scikit-learn. Finally, we embarked on learning PyTorch, a machine learning framework built on the Torch library. In this ninth article in the series on AI, we will focus on a topic that has not yet been covered -- natural language processing (NLP). For this purpose, we will utilise both PyTorch and the NLTK (natural language toolkit) libraries. We will also discuss PySpark, an interface for Apache Spark in Python. However, as always, we will start by solidifying our theoretical grasp of AI and machine learning.

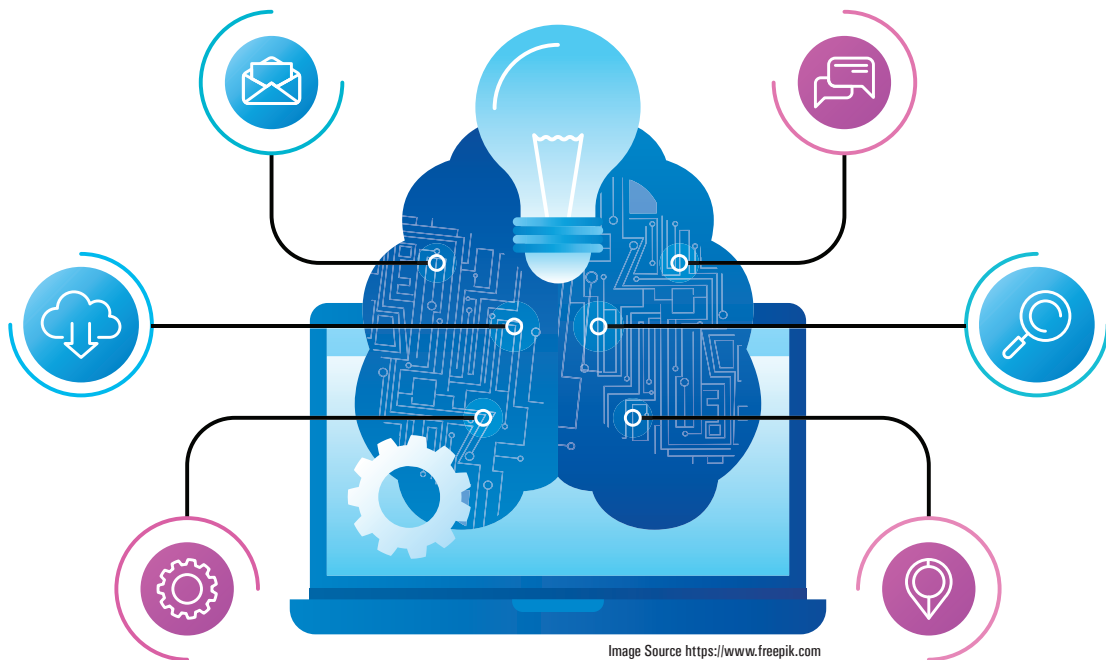


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Before we delve deeper into our discussion of neural networks, let us take a moment to consider the current state of affairs. If you examine the recent trends in the Indian engineering education sector, you will see a growing interest among young students to study computer science engineering. Although some tech companies have started layoffs, I believe the fascination with computer science engineering among the

younger generation will not decline in the near future. Of course, I may be wrong, and only time can tell. Now, what about the areas of interest among students who have recently begun studying computer science engineering? Many students today are interested only in fields like AI, machine learning, and data science. This choice is understandable, given the emergence of AI based tools like ChatGPT. The abundance of literature

in this field can make it difficult to keep up with the new terminology that is constantly being introduced. I recently participated in evaluating several undergraduate level academic projects based on AI and machine learning. The amount of new jargon presented by the project groups was both fascinating and concerning. Given that many of our bright young students plan to work in the fields of AI and machine learning in the near future,